Project Executable Files

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	**Item to Submit**	**Submitted (Yes / No / NA)**
1	Complete source code (all files and folders)	**Yes**
2	README / Setup Guide (instructions to run the project)	**Yes**
3	requirements.txt / package.json / dependency file	**Yes**
4	Database schema / seed files (if applicable)	**N/A**
5	Environment configuration file (.env.example or similar)	**N/A**
6	Deployed application URL (if hosted)	**Yes**
7	APK / Executable binary (if applicable for mobile/desktop apps)	**N/A**
8	Dockerfile / Containerization config (if applicable)	**N/A**

9	Test files and test results	**Yes**
10	Demo video or walkthrough (if required)	**Yes**

Step 2: File / Folder Structure

```
OptiCrop/
├── app.py
├── requirements.txt
├── README.md
├── .gitignore
├── dataset/
│   └── Crop_recommendation.csv
├── model/
│   ├── label_encoder.pkl
│   ├── model.pkl
│   ├── model_comparison.csv
│   └── scaler.pkl
├── notebooks/
│   └── ML_Model.ipynb
├── utils/
│   ├── helper.py
│   ├── predict.py
│   ├── preprocess.py
│   └── train.py
├── templates/
│   ├── about.html
│   ├── index.html
│   ├── layout.html
│   ├── predict.html
│   └── result.html
└── static/
    ├── css/
```

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│   └── style.css
├── js/
│   └── script.js
└── images/
    ├── confusion_matrix.png
    ├── feature_importance.png
    └── model_comparison.png
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://credit-card-approval-prediction-23p5.onrender.com
Login Credentials (Demo)	**Not Required (Application does not require authentication)**
Platform / Hosting Provider	**Render**
Repository Link	https://github.com/pmsaiakshay-ux/Smart-Agricultural-Production-Optimization-Engine.git
Demo Video Link	https://drive.google.com/file/d/12xoheUOexNiVmtcQ9c5jeOIC4LDttyFD/view?usp=sharin___g

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Step 4: Run Instructions

1. Clone the GitHub repository.
2. Navigate to the project folder.

3. git clone https://github.com/pmsaiakshay-ux/Smart-Agricultural-Production-Optimization-Engine.git
4. Create a virtual environment.

python -m venv venv

5. Activate the virtual environment.

Windows

venv\Scripts\activate

Linux / macOS
source venv/bin/activate

5. Install the required dependencies.

pip install -r requirements.txt

6. Ensure the dataset and trained model files are available in their respective folders.
7. Run the Flask application.

python app.py

8. Open your browser and visit:

http://127.0.0.1:5000

9. Enter applicant details and click **Predict Eligibility** to view the prediction result.

Step 5: Known Issues / Limitations

S.No	**Known Issue / Limitation**	**Workaround / Status**
1	The application currently uses a CSV dataset and serialized model files instead of a database.	Planned database integration in a future release.
2	User authentication and account management are not implemented.	Planned as a future enhancement.
3	The application is designed for small to medium workloads and has not been optimized for high concurrent user traffic.	Can be improved by using Docker, load balancing, caching, and cloud scaling.